

Complete Python Practical (NumPy + Pandas)

Covers Questions 1 to 7 (Exam Ready)

Q1: NUMPY

```
import numpy as np

# (a)
ARR1 = np.random.rand(3,4)
print(np.mean(ARR1, axis=1))
print(np.std(ARR1, axis=1))
print(np.var(ARR1, axis=1))

# (b)
m = int(input("m:"))
n = int(input("n:"))
arr = np.random.randint(1,100,(m,n))
print(arr.shape, type(arr), arr.dtype)
print(arr.reshape(n,m))

# (c)
arr1d = np.array([0,5,np.nan,3,0,np.nan])
print(np.where(arr1d==0))
print(np.where(arr1d!=0))
print(np.where(np.isnan(arr1d)))

# (d)
A1 = np.random.rand(5)
A2 = np.random.rand(5)
A3 = np.random.rand(5)
A4 = A3 - A2
A5 = 2*A1
print(np.cov(A1,A4))
print(np.corrcoef(A1,A5))

# (e)
A1 = np.random.randint(1,10,10)
A2 = np.random.randint(1,10,10)
print(np.sum(A1[:5]+A2[:5]))
print(np.prod(A1[5:]*A2[5:]))
```

```
# (f)
arr = np.random.rand(5,5)
print(arr.nbytes)

# (g)
m = int(input())
n = int(input())
arr = np.random.randint(10,100,(m,n))
r1 = int(input())
r2 = int(input())
arr[[r1,r2]] = arr[[r2,r1]]
col = int(input())
arr[:,col] = arr[:,col-1]
print(arr)
```

Q2: PANDAS SERIES

```
import pandas as pd
import numpy as np

# (a)
s = pd.Series([5,2,9,1,7], index=['a','e','c','b','d'])
print(s.sort_index())
print(s.sort_values())

# (b)
s2 = pd.Series(np.random.randint(1,5,10))
print(s2.rank(method='first').min())
print(s2.rank(method='max').max())

# (c)
print(s.idxmin())
print(s.idxmax())
```

Q3: DATAFRAME

```
import pandas as pd
import numpy as np

df = pd.DataFrame(np.random.rand(50,3), columns=['A','B','C'])
```

```

# insert NaN
for _ in range(int(df.size*0.1)):
    df.iat[np.random.randint(0,50), np.random.randint(0,3)] = np.nan

# (a)
print(df.isnull().sum())

# (b)
df = df.dropna(thresh=45, axis=1)

# (c)
df = df.drop(df.sum(axis=1).idxmax())

# (d)
df = df.sort_values(by=df.columns[0])

# (e)
df = df.drop_duplicates(subset=df.columns[0])

# (f)
print(df.iloc[:,0].corr(df.iloc[:,1]))
print(df.iloc[:,1].cov(df.iloc[:,2]))

# (g)
df['bins'] = pd.cut(df.iloc[:,1], bins=5)

```

Q4: MERGING DATAFRAMES

```

import pandas as pd

df1 = pd.DataFrame({'Name': ['A', 'B', 'C'], 'Date': [1,2,3], 'Duration': [30,40,50]})
df2 = pd.DataFrame({'Name': ['B', 'C', 'D'], 'Date': [1,2,3], 'Duration': [30,50,40]})

# (a)
print(pd.merge(df1, df2, on='Name'))

# (b)
print(pd.concat([df1, df2]).drop_duplicates(subset='Name', keep=False))

# (c)
merged = pd.concat([df1, df2])
print(len(merged))

```

```
# (d)
print(merged.set_index(['Name', 'Date']).describe())
```

Q5: IRIS DATA

```
from sklearn.datasets import load_iris
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['target'] = iris.target

# (a)
print(df.info())

# (b)
print(df.isnull().sum())

# (c)
df['target'].value_counts().plot(kind='bar')
plt.show()

# (d)
sns.regplot(x=df.iloc[:,2], y=df.iloc[:,0])
plt.show()

# (e)
df.iloc[:,3].plot(kind='density')
plt.show()

# (f)
sns.pairplot(df, hue='target')
plt.show()

# (g)
sns.heatmap(df.iloc[:,0:2].corr(), annot=True)
plt.show()

# (h)
print(df.describe())

# (i)
```

```
sns.heatmap(df.corr(), annot=True)
plt.show()
```

Q6: TITANIC

```
import seaborn as sns
import matplotlib.pyplot as plt

df = sns.load_dataset('titanic')

# (a)
df = df.drop(columns=df.isnull().sum().idxmax())

# (b)
print(len(df[df['age']>30]))

# (c)
print(df[df['pclass']==2]['fare'].sum())

# (d)
print(df.groupby('pclass')['survived'].sum())

# (e)
print(df.groupby('sex')['age'].describe())

# (f)
sns.scatterplot(data=df, x='fare', y='age', hue='sex')
plt.show()

# (g)
sns.kdeplot(df['age'])
sns.kdeplot(df['fare'])
plt.show()

# (h)
df['pclass'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.show()

# (i)
print(df.groupby('pclass')['survived'].mean()*100)
```

Q7: FAMILY INCOME

```
import pandas as pd

data = {
    'FamilyName':
    ['Shah', 'Vats', 'Vats', 'Kumar', 'Vats', 'Kumar', 'Shah', 'Shah', 'Kumar', 'Vats'],
    'Gender':
    ['Male', 'Male', 'Female', 'Female', 'Female', 'Male', 'Male', 'Female', 'Female', 'Male'],
    'Income': [44000, 65000, 43150, 66500, 255000, 103000, 55000, 112400, 81030, 71900]
}

df = pd.DataFrame(data)

# (a)
print(df.groupby('FamilyName')['Income'].sum())

# (b)
print(df.groupby('FamilyName')['Income'].agg(['max', 'min']))

# (c)
print(df[df['Income'] < 80000])

# (d)
females = df[df['Gender']=='Female']
print(len(females))
print(females['Income'].mean())

# (e)
avg = df['Income'].mean()
df = df[df['Income'] >= avg]
print(df)
```

End of File

(Ready for exam writing / PDF conversion)